

Computer

The History and future of technology

History of Computer

- The computer start in 19th century and went on change the world during the 20th century.

1st modern computer

- The 1st computer **Atanasoff–Berry** and his graduate student Clifford Berry **computer** was invented by **Atanasoff** in 1942.
- It was made up of vacuum tubes and its weighed 700 pounds. Atanasoff was credited with several breakthrough ideas that are still present in modern computers.
- It was basis for the modern computer we all use today.
- **Atanasoff** was credited with several breakthrough ideas that are still present in modern computers.

1st generation of computer

- The period of first generation was **from 1940-1956** by **Charles Babbage**.
- The computers of first generation used vacuum tubes as the basic components for memory and circuitry for CPU (Central Processing Unit). These tubes, like electric bulbs, produced a lot of heat and the installations used to fuse frequently.
- The main features of 1st gen computer is;
- Vacuum tube technology
- Unreliable
- Supported machine language only
- Very costly
- Generated a lot of heat
- Slow input and output devices
- Huge size
- Need of AC
- Non-portable
- Consumed a lot of electricity

2nd generation of computer

- The 2nd gen computer was invented Bell Labs by **Walter H. Brattain (1902-1987), John Bardeen (1908-1991), and William B.**
- The 2nd gen computer was made upon transistors. transistors were used that were cheaper, consumed less power, more compact in size, more reliable and faster.
- In this generation, assembly language and high-level programming languages like FORTRAN, COBOL were used.
- The main features of 2nd gen is;
- Use of transistors
- Reliable in comparison to first generation computers
- Smaller size as compared to first generation computers
- Generated less heat as compared to first generation computers
- Consumed less electricity as compared to first generation computers
- Faster than first generation computers
- Still very costly
- AC required
- Supported machine and assembly languages

3rd Generation of computer

- The period of third generation was from 1965-1971. The computers of third generation used Integrated Circuits (ICs) in place of transistors. A single IC has many transistors, resistors, and capacitors along with the associated circuitry.
- The IC was invented by **Jack Kilby**.
- In this generation High-level languages (FORTRAN-II TO IV, COBOL, PASCAL PL/1, BASIC, ALGOL-68 etc.)
- The main features of third generation are –
 - IC used
 - More reliable in comparison to previous two generations
 - Smaller size
 - Generated less heat
 - Faster
 - Lesser maintenance
 - Costly
 - AC required
 - Consumed lesser electricity
 - Supported high-level language